

INTRODUCTION TO COMPUTER GRAPHICS

INSTRUCTOR:

DOÇ. DR. UFUK ÇELİKCAN

INSTRUCTOR

Office: Room 204

ufuk.celikcan@gmail.com for class-related e-mail exchange

OVERVIEW

Book:

Ed Angel and Dave Shreiner,

**Interactive Computer Graphics, A Top-down
Approach with OpenGL (7th Edition)**

Addison-Wesley

OBJECTIVES

Broad introduction to Computer Graphics

- Software
- Hardware
- Applications

Top-down approach

Shader-Based WebGL

PREREQUISITES

Some programming skills

Basic Data Structures

- Linked lists
- Arrays

Geometry

Simple Linear Algebra

412 GRADING

Quizzes: 20% (~10 quizzes)

Exam: 40% (near the end of the term)

Final Project (details later): 40%

412 GRADING

Exam:

Questions similar to exercises done in class

+

Shader-coding questions

**(not unlike the ones you will be submitting for the
BBM414 Lab assignments)**

BBM412 GRADING

- A project implemented in Javascript (+some basic HTML) and GLSL with Shader-Based WebGL is presented at the end of the class.

https://docs.google.com/document/d/e/2PACX-1vQcrgEjFGbnHcks_3WQEtgcZNV5CVBa-dn8TEVMM6wBLYycm4DCypIHF6W0ugrd99St_JVzmKrDG2RZ/pub

- Each project is realized by a team of 3 students (*or 4).
- The finalized project is required to exhibit the knowledge of an array of subjects covered during the class.
- The submission for the project includes:
 - the entire project directory including all source files and accessory files
 - a video teaser/trailer of the finalized project
 - and the powerpoint presentation of the project.

BBM414 GRADING

- 5 Lab Sessions: Each session includes an in-lab assignment (16% each).
- 1 Project Assignment: Worth 20%.
- In-lab assignments are completed during the scheduled lab under the supervision of your BBM414 TA and must be submitted before the lab session ends. Your TA will provide guidance and tips as needed.
- All assignments are individual work.
- Assignments will be implemented in Javascript (+some basic HTML) and GLSL with Shader-Based WebGL.
- To avoid receiving an F1 grade, you must submit at least 4 out of 5 in-lab assignments.

WEB RESOURCES

- https://www.cs.unm.edu/~angel/BOOK/INTERACTIVE_COMPUTER_GRAPHICS/SEVENTH_EDITION/
- <http://interactivecomputergraphics.com/> (code for the 8th edition but still can be helpful)

OUTLINE: PART 1

Part 1: Introduction

Text: Chapter 1

- What is Computer Graphics?
- Applications Areas
- History
- Image formation
- Basic Architecture

OUTLINE: PART 2

Part 2: Basic OpenGL

Text: Chapter 2

- Architecture
- Simple programs in two and three dimensions
- Basic shaders and GLSL
- Interaction

OUTLINE: PART 3

Part 3: Three-Dimensional Graphics

Text: Chapters 3-5

- Geometry
- Transformations
- Homogeneous Coordinates
- Viewing
- Lighting and Shading

OUTLINE: PART 4

Part 4: Implementation

Text: Chapter 6

- Approaches (object vs image space)
- Implementing the pipeline
- Clipping
- Line drawing
- Polygon Fill
- Display issues (color)

OUTLINE: PART 5

Part 5: Discrete Methods

Text: Chapter 7

- Buffers
- Pixel Maps
- Texture Mapping
- Compositing and Transparency

WHAT IS COMPUTER GRAPHICS?

LECTURER:

DR. UFUK ÇELİKCAN

OBJECTIVES

- In this lecture, we explore what computer graphics is about and survey some application areas
- With a historical introduction

COMPUTER GRAPHICS

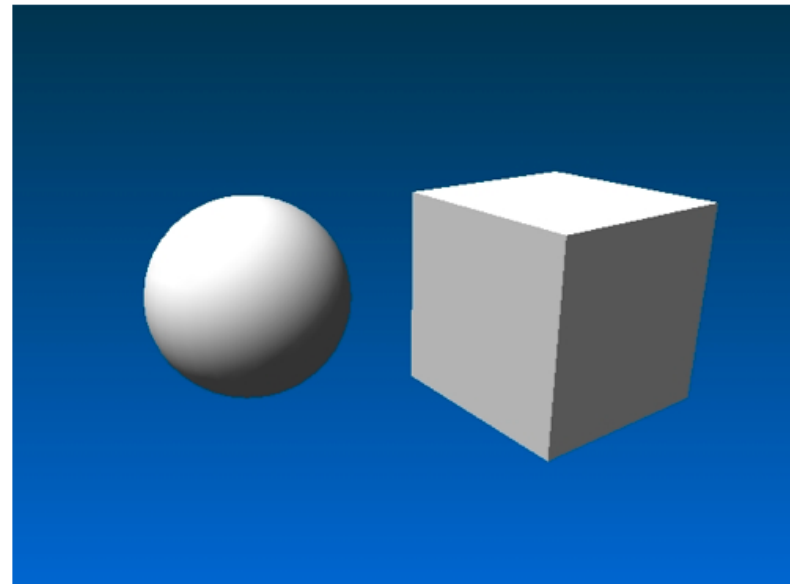
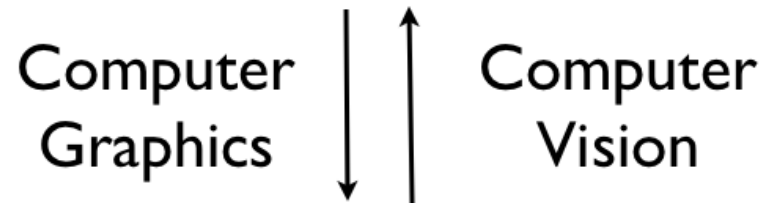
Computer graphics is concerned with all aspects of creating visuals with a computer

- Hardware
- Software
- Applications

COMPUTER GRAPHICS

`(cube, size, x0, y0, z0,  $\theta_{xy}$ ,  $\theta_{xz}$ ,  $\theta_{yz}$ , ...)`
`(sphere, radius, x1, y1, z1, ...)`

- Computer Graphics algorithms go from the parameter space to the image space (**rendering**)
- Computer Vision algorithms do the opposite (inverse-rendering).



COMPUTER GRAPHICS

The applications of computer graphics are many and varied; we can, however, divide them into four major areas:

1. Display of information
2. Design
3. Simulation and animation
4. User interfaces

Although many applications span two or more of these areas, the development of the field was based on separate work in each.

EXAMPLE

Where did this image come from?



What hardware/software did we need to produce it?

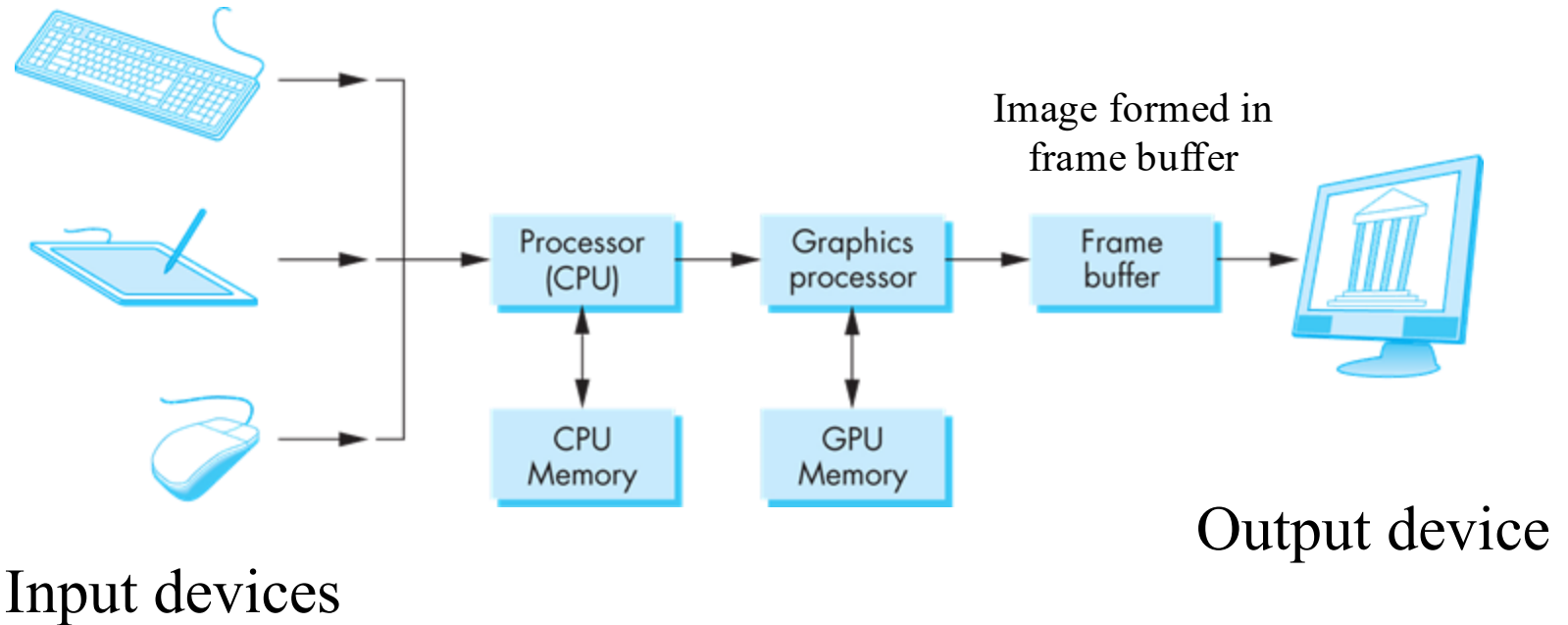
PRELIMINARY ANSWER

- **Application:** The object is an artist's rendition of the sun for an animation to be shown in a domed environment (planetarium)
- **Software:** Maya for modeling and rendering, Maya is built on top of OpenGL
- **Hardware:** Computer with graphics card for modeling and rendering



BASIC GRAPHICS SYSTEM

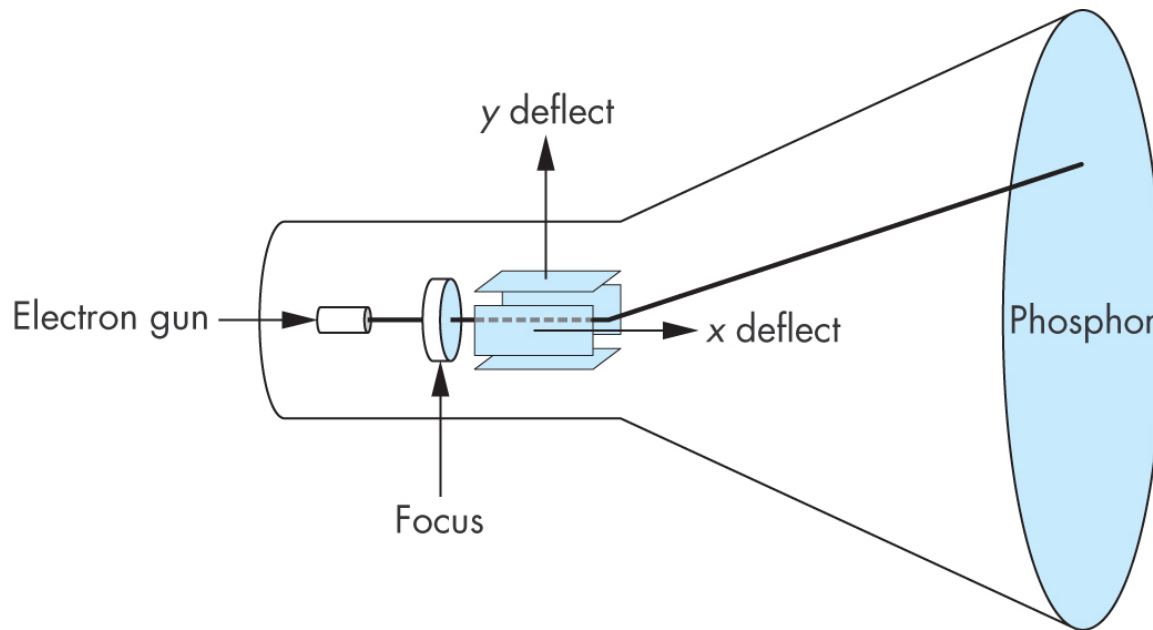
A computer graphics system is basically a computer system; as such, it must have all the components of a general-purpose computer system.



This model is general enough to include workstations and personal computers, interactive game systems, mobile phones, GPS systems, and sophisticated image generation systems.

CATHOD RAY TUBE (CRT)

- The field of Computer Graphics began humbly in 1950s
 - with the display of a few lines on a cathode-ray tube (CRT);
- Until 2000s, CRT was the dominant type of display (or monitor)



Can be used either as a line-drawing device (calligraphic) or to display contents of frame buffer (raster mode)

COMPUTER GRAPHICS: 1950-1960

Computer graphics goes back to the earliest days of computing

- Strip charts



- Pen plotters



- Simple displays using A/D converters to go from computer to calligraphic CRT

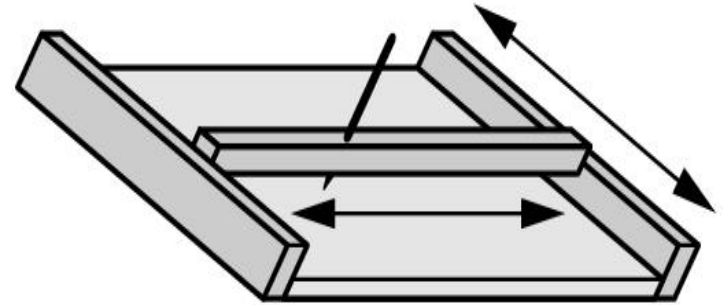


Cost of refresh for CRT too high

- Computers slow, expensive, unreliable

PEN-PLOTTER MODEL

- Historically, most early graphics systems were only two-dimensional. The conceptual model that they used is now referred to as the pen-plotter model, referencing the output device that was available on these systems



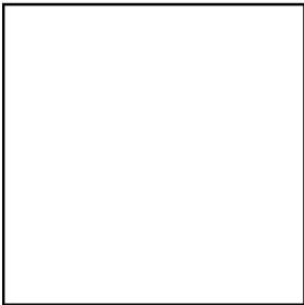
Typical example of “sequential access”

Two basic functions for drawing:

`moveto(x , y) – pen up`

`lineto(x , y) – pen down`

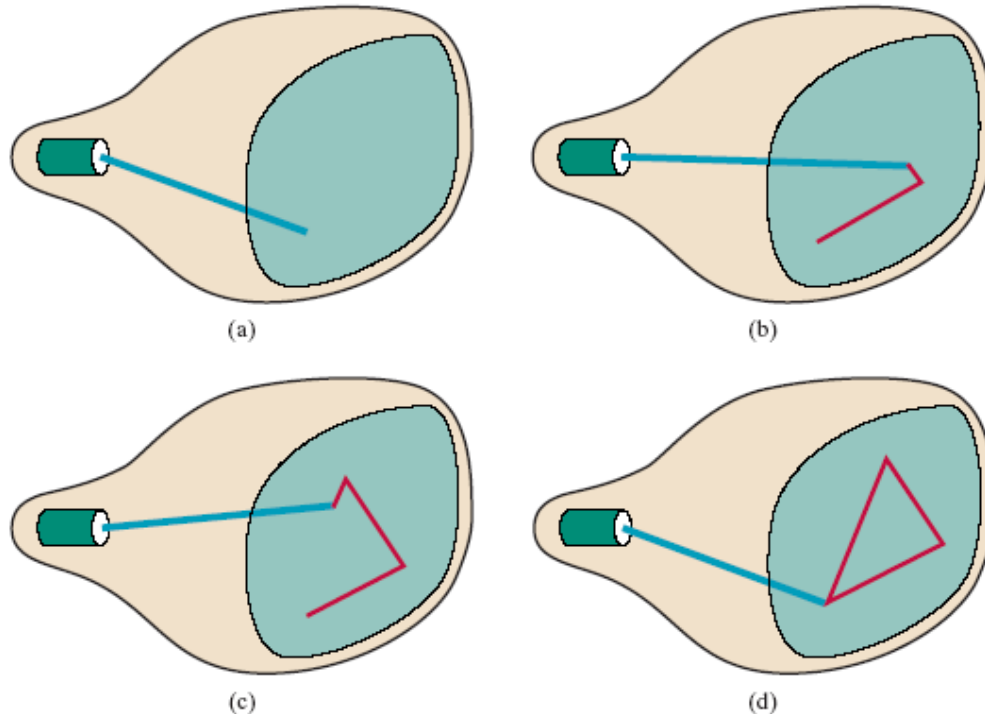
```
moveto (0,0);  
lineto(1,0);  
lineto(1,1);  
lineto(0,1);  
lineto(0,0);  
{ draws a square }
```

A simple 2D coordinate system with a square drawn. The square's vertices are at (0,0), (1,0), (1,1), and (0,1). The code to the left of the square describes the sequence of pen movements to draw this square.

- Various APIs—such as LOGO and PostScript—have their origins in this model.
- Although they differ from one another, they have a common view of the process of creating an image as being similar to the process of drawing on a pad of paper.

VECTOR (CALLIGRAPHIC / RANDOM SCAN) DISPLAYS

- stored as a set of line-drawing commands in an area of memory (refresh display file, display list, display program)
- draw a picture one line at a time



VECTOR (CALLIGRAPHIC / RANDOM SCAN) DISPLAYS

The vector games from that era included:

- Asteroids
- Battlezone
- Lunar Lander
- Star Trek
- Star Wars
- Tempest



VECTOR (CALLIGRAPHIC / RANDOM SCAN) DISPLAYS

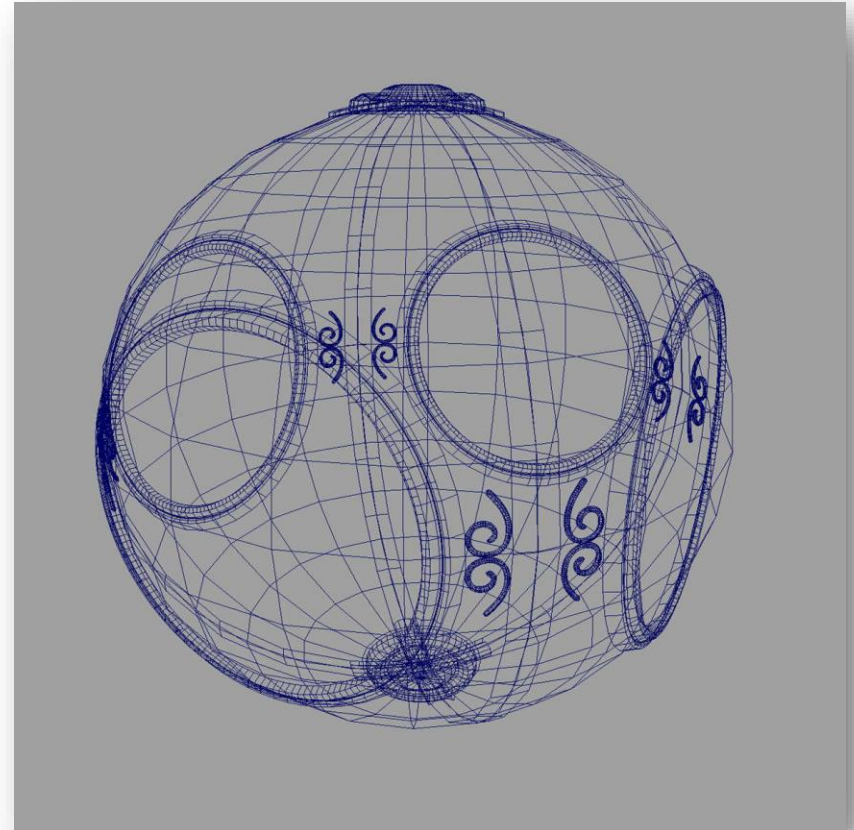
Atari Star Wars Arcade (1983)



COMPUTER GRAPHICS: 1960-1970

- **Wireframe graphics** - a 3d vectograph, showing the object rendered using only line segments.
- **Sketchpad**
- **Display Processors**

wireframe representation
of “the sun” object



SKETCHPAD

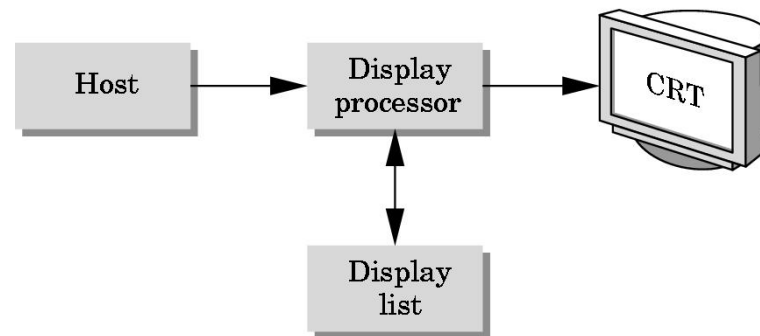
Ivan Sutherland's PhD thesis at MIT

- Recognized the potential of man-machine interaction
- Loop
 - Display something
 - User moves light pen
 - Computer generates new graphic
- Sutherland also created many of the now common algorithms for computer graphics



DISPLAY PROCESSOR

- Rather than have the main processor try to refresh display, use a special purpose processor called a *display processor* (DPU)



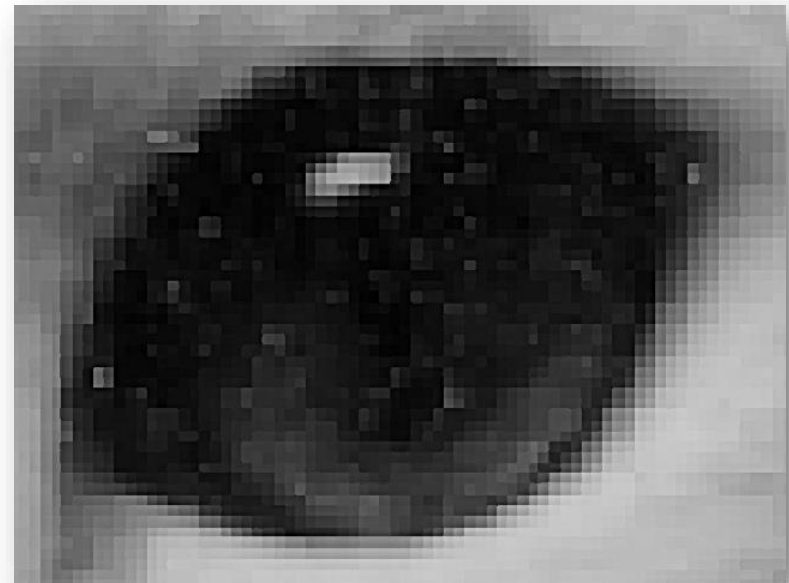
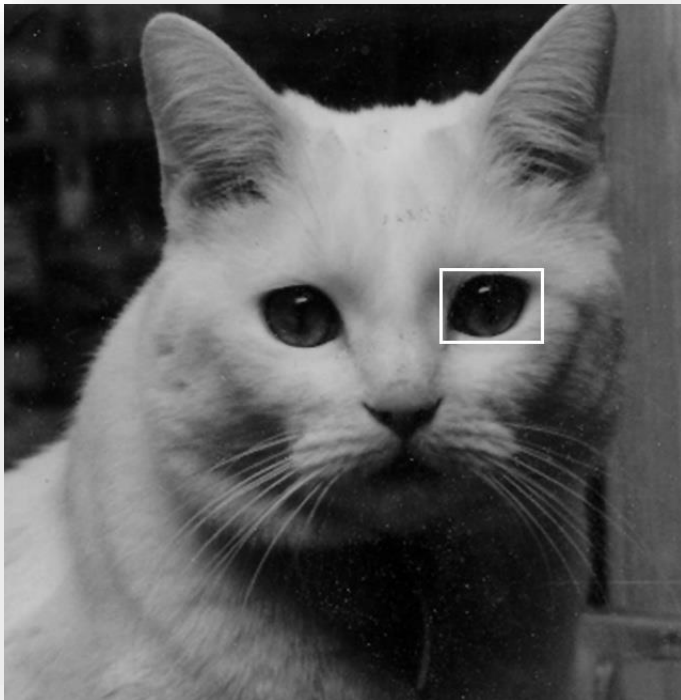
- Host *compiles* display list and sends to DPU
- Graphics stored in display list (display file) on display processor

COMPUTER GRAPHICS: 1970-1980

- **Raster Graphics**
- **Beginning of graphics standards**
 - IFIPS
 - GKS: European effort
 - Becomes ISO 2D standard
 - Core: North American effort
 - 3D but fails to become ISO standard

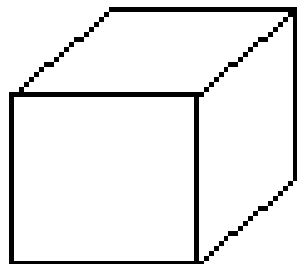
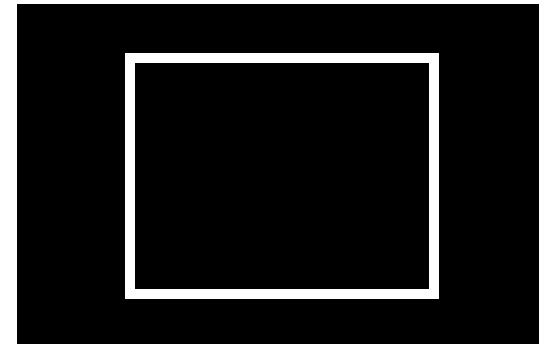
RASTER GRAPHICS

- Virtually all modern graphics systems are raster based.
- The image we see on the output device is an array—the **raster**—of picture elements (**pixels**) produced by the graphics system.
- Collectively, the pixels are stored in a part of memory called the **frame buffer**.
 - Its resolution—the number of pixels in the frame buffer— determines the detail that you can see in the image.
 - In a very simple system, the frame buffer holds the colored pixels that are displayed on the screen. In most systems, it holds far more information, including the depth information to create images from three-dimensional data.



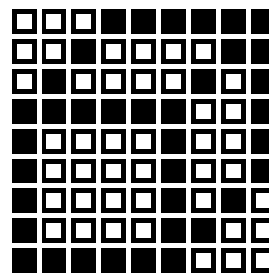
RASTER GRAPHICS

0	0	0	0	0	0	0	0	0	0	0	0	0	0
0	0	1	1	1	1	1	1	1	1	0	0	0	0
0	0	1	0	0	0	0	0	0	1	0	0	0	0
0	0	1	0	0	0	0	0	0	1	0	0	0	0
0	0	1	0	0	0	0	0	0	1	0	0	0	0
0	0	1	0	0	0	0	0	0	1	0	0	0	0
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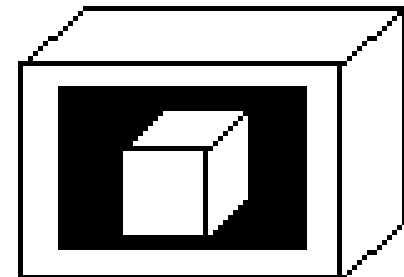
model

graphics
software



frame buffer

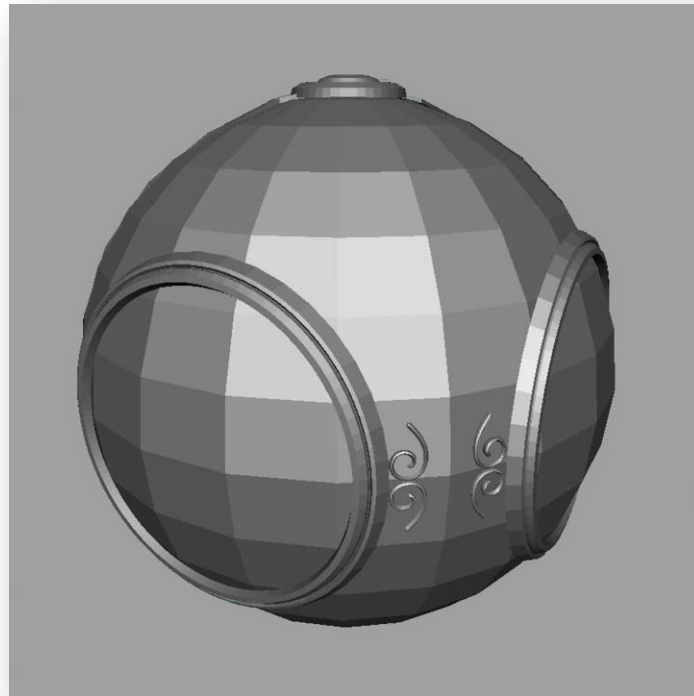
graphics
hardware



monitor screen

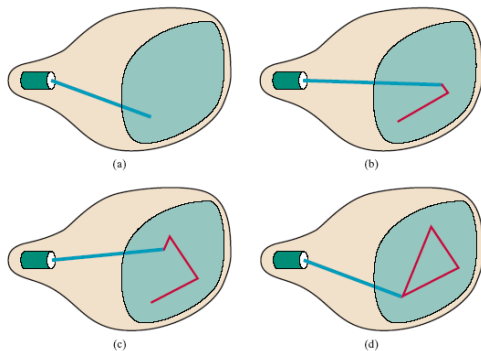
RASTER GRAPHICS

Allows us to go from lines and wire frame images to filled polygons with facet shading, which simulates how light reflects off surfaces.



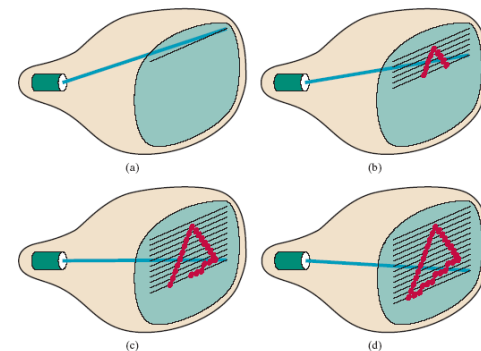
VECTOR GRAPHICS

- lines drawn directly, no predefined grid
- commands tell the electron gun where to move and when to turn on/off
- + lines are smooth
- + close to the 'pure mathematics' of the model
- slower with more elements to be drawn, can start to flicker
- only lines possible, no filled polygons, or bitmaps
- monochrome for each electron gun



RASTER GRAPHICS

- image represented by a rectangular grid of pixels and stored in a frame buffer
- electron gun(s) continually scanning in a regular pattern (line by line across entire screen)
- + constant time to redraw any number of elements
- + no flicker
- at low resolutions: jagged edges visible due to conversion to pixels
- discretized version of the model



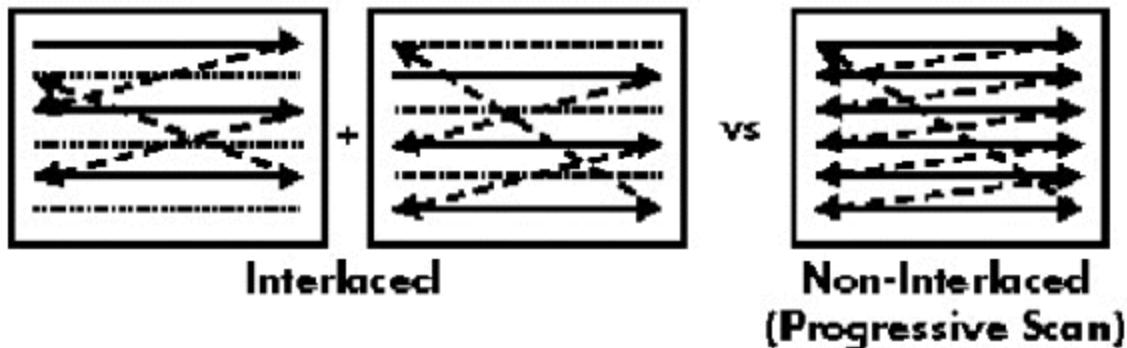
RASTER GRAPHICS

Retrace

- Horizontal retrace/Vertical retrace

Interlaced refresh procedure

- used with slower refreshing rates
 - avoid flicker
 - provide adjacent scan lines



COMPUTER GRAPHICS: 1980-1990

- **Special purpose hardware**
 - **Silicon Graphics** revolutionized the graphics workstation with the *geometry engine* (1982)
 - VLSI implementation of *graphics pipeline* in a special-purpose chip
 - To use the system, application programmers used a library called **GL**. With GL, it was relatively simple to program three dimensional interactive applications
- **Industry-based standards**
 - PHIGS (Programmer's Hierarchical Interactive Graphics System)
 - Pixar's RenderMan



COMPUTER GRAPHICS: 1980-1990

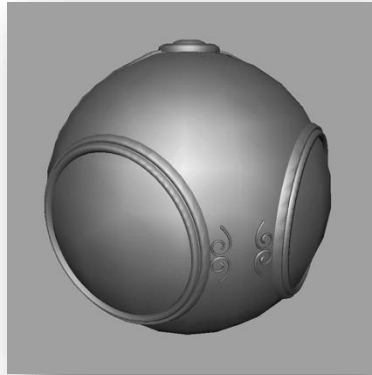


Human-Computer Interface (HCI) : GUI

- **1984 – Apple Macintosh popularizes the GUI. Super Bowl commercial shown once, most expensive ever made at that time**
- **1984 – Networked graphics: MIT's X Window System: hardware- independent platform and networking protocol for developing GUIs on UNIX-like systems**
- **1985 – Windows 1.0 – provided GUI interface to MS-DOS. No overlapping windows (tiled instead).**

COMPUTER GRAPHICS: 1980-1990

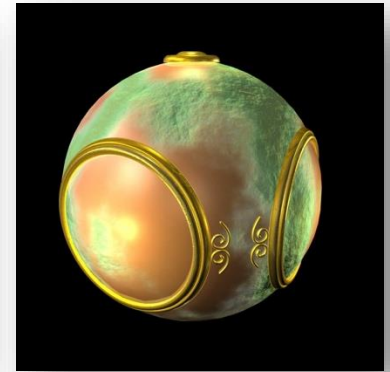
Realism comes to computer graphics



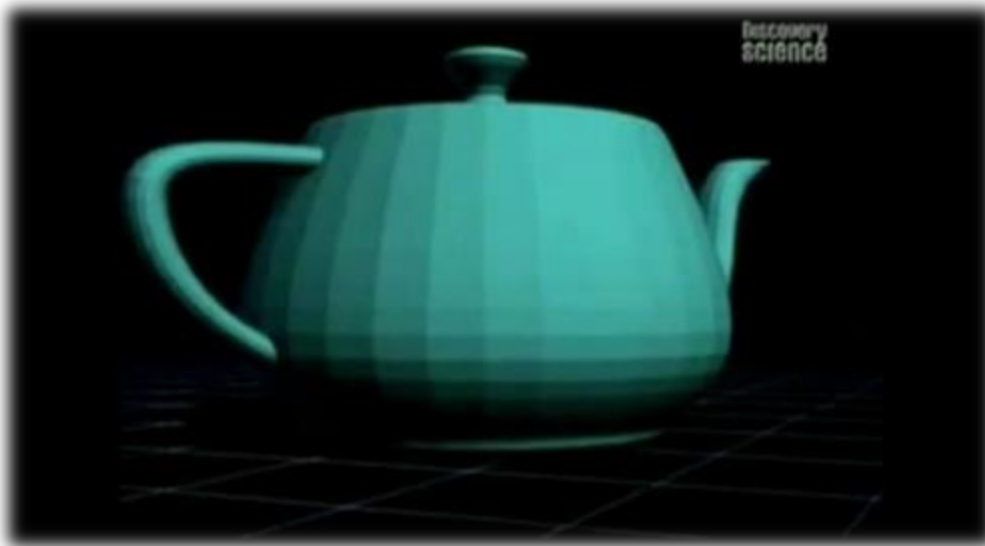
smooth shading



environment mapping



bump mapping



Discovered the key element in making the object appear more realistic was **lighting**

Facets are smoothed over to make the shading look continuous

Next the computer simulates the highlights figuring out where they should show up and how bright they should be knowing where the light sources in the scene are

COMPUTER GRAPHICS: 1990-2000

- **GL led to OpenGL API (1992)**
 - a platform-independent API that is
 - Easy to use
 - Close enough to the hardware to get excellent performance
 - Focus on rendering
 - Omitted windowing and input to avoid window system dependencies
- **Completely computer-generated feature-length movies (*Toy Story* (1995)) are successful**
- **New hardware capabilities**
 - Texture mapping
 - Blending
 - Accumulation, stencil buffers



COMPUTER GRAPHICS: 2000-

- **Graphics cards for PCs dominate market**

- Today, virtually all graphics systems are characterized by special-purpose graphics processing units (GPUs)
 - NVidia, ATI
- GPUs have evolved to where they are as complex or even more complex than CPUs.
- They are characterized by both special-purpose modules geared toward graphical operations and a high degree of parallelism
- Modern GPUs contain 1000s of processing units, each one user programmable.
- GPUs are so powerful that they can be used as mini supercomputers for general purpose computing.



COMPUTER GRAPHICS: 2000-

- **Programmable pipelines**
 - Major advance in pipeline architecture: vertex and the fragment processor are now both programmable by the application program.
 - many techniques that could not be executed in real time with the fixed-function pipeline are now possible in real time.
 - Bump mapping is one example that is now programmable but formerly could only be done off-line.



- **Game boxes and game players determine direction of market**
- **Computer graphics routine**
- **in movie industry**
- **Photorealism**

